

UNNI ADITYA MOHAN

Boston, MA | (857) 225-3866 | adityamohan2604@gmail.com | linkedin.com/in/adityamohan2604 | github.com/adityamohan260498

SUMMARY

Mechanical engineer with 3 years of capital-equipment experience at Thermax on boiler pressure parts, plus a simulation background across transient heat transfer, multiphase CFD and structural FEA. Comfortable owning a design from CAD through analysis to procurement and the manufacturing floor.

EDUCATION

Northeastern University, Boston, MA · Sep 2024 – Present

Master of Science in Robotics · GPA 3.82

National Institute of Technology Rourkela, India · Aug 2017 – Jun 2021

B.Tech in Mechanical Engineering, with Honors · CGPA 8.12/10 · Coursework: Thermal Engineering, Heat Transfer, Fluid & Solid Mechanics, CAD, CFD

TECHNICAL SKILLS

- **CAD & Drafting:** SolidWorks, AutoCAD (2D/3D)
- **Simulation:** ANSYS Mechanical (Static Structural, Explicit Dynamics, Topology Optimization), ANSYS Fluent (CFD, VOF multiphase), COMSOL Multiphysics (conjugate heat transfer, porous media)
- **Engineering Analysis:** Heat transfer & thermal management, pressure-part design, structural FEA, fin and heat-exchanger sizing, psychrometrics, design trade studies
- **Programming & Tools:** MATLAB & Simulink, Python, C++, ROS 2, OpenCV, Arduino

PROFESSIONAL EXPERIENCE

Thermax Ltd, Pune, India · Sep 2021 – May 2024

Project Engineer

- Primary coordinator between engineering, procurement and manufacturing on capital boiler and pressure-part projects; identified critical-path delays in design approval and implemented adjustments that recovered **2 weeks** of delivery schedule.
- Designed a reusable transportation frame for pressure parts in AutoCAD, saving **\$20K per project**.
- Built and trained a machine learning model predicting optimal steam drum design parameters, automating the CAD design process, saving **20 man-hours** and improving dimensional accuracy.

ENGINEERING PROJECTS

Thermal Modelling of a Metal Hydride Hydrogen Storage Reactor · Dec 2020 – May 2021

B.Tech Thesis · COMSOL Multiphysics

- Built a transient model coupling the energy equation, Darcy flow through a porous bed and Arrhenius kinetics for a 30 bar hydrogen storage vessel with 36 embedded cooling tubes and internal copper fins.
- Ran 13 transient studies across fin count (4–10), fin thickness (1–3 mm) and shell-side convection (100–2000 W/m²K), cutting time to 90% charge from **147 s to 88 s (-40%)**.
- Selected a 10 × 1 mm fin array hitting that target with **38% less parasitic copper mass** than the equal-speed alternative; a shell-side *h* sweep moved charge time only 1.1%, deleting a forced-convection jacket from the design.

Design and Analysis of a Humidity / Climate Test Chamber · 2020 – Mar 2021

Research Internship · SolidWorks, ANSYS Fluent

- Designed a two-unit environmental test chamber for 85/85 damp-heat and HALT/HASS qualification: a cross-flow finned heat exchanger setting dry-bulb temperature, feeding an atomizer humidification stage.
- Conjugate heat transfer analysis gave **30 to 23 °C per stage** on 0 °C chilled water, specifying three stages in series; air-side streamlines (peak 5.20 m/s) identified fin pitch as the limiting resistance.
- Multiphase VOF study showed the atomizer discharging a coherent liquid core (peak volume fraction **0.968**) trapped in a vortex over the first 30% of chamber length, driving nozzle placement and baffling.

Mechanical Design Lead · Autonomous Underwater Vehicle (Team of 6) · Jan 2018 – Mar 2021

Built for ROBOSUB, SAUVC and SAVe competitions

- Achieved **42% drag reduction** and improved stability through hull redesign and CFD in ANSYS Fluent.
- Topology optimization in ANSYS Mechanical cut chassis weight **20%** while holding structural integrity at 15 ft depth (**1.8× safety factor**).
- Developed sealed subassemblies for fully submerged service: a piston ballast tank reaching full buoyancy in under 10 s at 15 ft, a torpedo launcher with 3–5 m precision, and a gripper rated to 1 kg underwater.

Design Lead · SAE Effi-Cycle Chassis and Suspension (Team of 4) · Sep 2017 – Apr 2020

Human-electric hybrid vehicle, SAE India

- Led design of a competition-compliant **15 kg** chassis and suspension for a two-passenger solar-assisted vehicle, from CAD through fabrication.
- Ran FEA in ANSYS Workbench (Static Structural, Explicit Dynamics) across front impact, side impact, rollover and torsion; all four passed at a minimum factor of safety of **1.24**.

ISS Active Thermal Control System · ROS 2 and Gazebo · Dec 2024 – Feb 2025

Space Station OS · open-source collaboration with JAXA

- Implemented a two-stage active thermal control system in C++ (internal water loops to an external ammonia loop and radiators); URDF-derived lumped-capacitance network integrated with RK4, with orbital solar input from an ECI sun vector peaking at **13.9 kW** across six panels.

ACHIEVEMENTS

- **Winners** · SAVe 2019, IIT Madras, India | **Semi-Finalists** · ROBOSUB 2019, San Diego, USA